Effect of regional anesthesia timing on clinical outcomes in isolated flail chest: A nationwide analysis

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BACKGROUND: Flail chest can cause pain and respiratory failure. The impact of early initiation of regional anesthesia (RA) in patients with flail chest remains unclear. We evaluated whether earlier RA (≤ 24 h) in isolated flail chest improves outcomes.

METHODS: We conducted a retrospective cohort study using data from the ACS-TQIP (2017–2023). Adults (≥ 18 y) with isolated flail chest receiving treatment were included. We excluded penetrating trauma, extra-thoracic Abbreviated Injury Scale ≥ 3, discharged within 24 hours, transfers, emergency department deaths, pre-hospital arrest, and pre-hospital or emergency department intubation. Cases were divided into early RA (≤ 24 h) and late RA (> 24 h). The primary outcome was unplanned intubation, and the secondary outcomes included intensive care unit (ICU) admission, discharge disposition, hospital, and ICU length of stay. Multivariable regression adjusted for patient characteristics, thoracic Abbreviated Injury Scale, mechanism of injury, surgical stabilization of rib fractures (SSRF), modified frailty index (mFI-11), pre-injury anticoagulation, and shock index. A subanalysis evaluated the SSRF, nonoperative management cohort, and RA technique.

RESULTS: There were 4,563 isolated flail chest cases with a median age of 59 years. 72.5% were male, and most were motor vehicle collisions (44.6%). 24.2% of patients received early RA. Late RA was associated with increased odds of unplanned intubation [adjusted odds ratio (aOR), 1.52; 95% CI, 1.09–2.12, $p = 0.015$], unplanned ICU admission (aOR, 1.93; 95% CI, 1.34–2.79, $p < 0.001$), longer hospital and ICU stays ($p < 0.001$), and lower odds of routine discharges (aOR, 0.83; 95% CI, 0.70–0.99, $p = 0.036$). In the SSRF cohort, early administration of RA was associated with shorter hospital and ICU stays. In the nonoperative management cohort, early RA was associated with more routine discharges, fewer ICU admissions, and shorter hospital and ICU stays.

CONCLUSIONS: In isolated flail chest, early RA is associated with lower rates of unplanned intubations and ICU admissions, shorter hospital and ICU stays, and higher rates of routine discharge. These findings support prioritizing early RA intervention, which is associated with favorable outcomes. (*J Trauma Acute Care Surg* 2026;00:000–000 Copyright © 2026 Wolters Kluwer Health, LLC. All rights reserved.)

LEVEL OF EVIDENCE: Retrospective Study, Therapeutic/Care Management; Level III.

KEY WORDS: Flail chest, regional anesthesia, thoracic trauma.

One of the most severe manifestations of blunt thoracic trauma is flail chest, which is associated with significant morbidity and mortality. Flail chest comprises 6% of all blunt chest trauma, with an in-hospital mortality rate up to 33%.[1] Patients with flail chest have high rates of complications such as respiratory failure, prolonged ventilation, pneumonia, and increased length of stay (LOS) in comparison to those without flail chest.[2] Over the past decade, surgical stabilization of rib fractures (SSRFs) has been increasingly used in flail chest cases and is associated with decreased complications and LOS.[3] Despite growing interest in SSRF and expanding supporting literature, operative stabilization is still not uniformly used in patients with flail chest. Hence, the use of analgesia in this population is crucial.[2,4]

Pain management is the mainstay of treatment for rib fractures because pain can limit ventilation and increase the risk of impaired secretion clearance, atelectasis, and infection. Regional anesthesia (RA), such as paravertebral blocks, thoracic epidural analgesia, and plexus blocks, is widely used because it improves respiratory mechanics and reduces pulmonary complications in these patients.[5–7] Some studies suggest that although RA is commonly used in rib fractures and flail chest, its impact on LOS, intubation, and intensive care unit (ICU) admission could potentially be confounded by indication and patient selection.[8]

Despite the crucial role of RA in flail chest management, data addressing this cohort are limited. Most

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studies combine management of flail chest and rib fractures and focus primarily on SSRF rather than on analgesic strategies.[9] Many landmark studies on RA in chest wall trauma predate modern trauma care, and now RA is increasingly incorporated into the management plan. Recent literature supports the potential benefit of improving analgesia, as it reduces respiratory complications and resource utilization, highlighting the importance of reassessing the timing of RA in flail chest management.[10–12] The timing of RA is not well-defined; therefore, we aimed to evaluate the association between early anesthesia (EA) and late anesthesia (LA) in patients with isolated flail chest to compare clinical outcomes. We hypothesized that EA would be associated with better outcomes and decreased resource utilization.

METHODS

We conducted a retrospective cohort study using the American College of Surgeons Trauma Quality Improvement Program (ACS-TQIP) from 2017 to 2023. ACS-TQIP is a national database that includes data from over 900 trauma centers across the United States. At these centers, trained registrars collect data in accordance with the National Trauma Data Standard guidelines.[13] This registry is a tool to improve the quality of trauma care and support research to improve patient outcomes. Since this database has de-identified data, this retrospective study was exempt from patient consent after institutional review board review and approval. The Strengthening the Reporting of Observational Studies in Epidemiology guidelines were used to ensure appropriate development and documentation of this study (Supplemental File 1, Supplemental Digital Content 1, http://links.lww.com/TA/F580).

Inclusion and Exclusion Criteria

We included all adult patients (≥ 18 y) with isolated flail chest who received RA. RA was defined as the administration of epidural analgesia, muscle, plexus, nerve block, or cryoablation. Flail chest was identified using Abbreviated Injury Scale (AIS) codes. Flail chest was identified using AIS diagnosis codes, which are defined as three or more contiguous rib fractures in two or more places. The TQIP database lacks variables that document clinical findings, so we could not confirm whether each case represented a clinically apparent flail chest or just a radiographic flail (Appendix A: Table 1; Supplemental File 2, Supplemental Digital Content 2, http://links.lww.com/TA/F581). Patients receiving RA were identified using the International Classification of Diseases ICD-10 procedure codes (Appendix A: Table 1; Supplemental file 2, Supplemental Digital Content 2, http://links.lww.com/TA/F581). We excluded patients with penetrating trauma, extra-thoracic AIS ≥ 3, missing RA timings, emergency department (ED) death, discharged in 24 hours, interfacility transfers, or pre-hospital or ED intubation.

Data Points and Grouping

We collected patient information, including age, gender, race, ED vital signs such as systolic blood pressure (SBP), heart rate (HR), and Shock Index (SI), mechanism of injury, Injury Severity Score, thoracic AIS, and rib fixation. Patients were grouped into EA and LA, defined as ≤ 24 hours and > 24 hours from ED presentation, respectively.

Outcomes

The primary outcome in this study was unplanned intubation. The secondary outcome included in-hospital complications, length of hospital and ICU stay, ventilator days, and discharge disposition. In-hospital complications assessed were deep vein thrombosis, pulmonary embolism, acute respiratory distress syndrome, ventilator-associated pneumonia, unplanned return to operating room, and unplanned ICU admission. The discharge disposition encompasses routine discharge, inpatient rehabilitation, skilled nursing facility, long-term acute care hospital, and other categories, which include intermediate care facility, short-term general hospital, hospice, home health services, psychiatric hospital, court or law enforcement, or discharge against medical advice.

Data Reporting and Statistical Analysis

We compared the baseline characteristics and outcomes of patients undergoing early and late RA. In our primary analysis, we performed a multivariable logistic regression to evaluate the association between anesthesia timing and relevant clinical outcomes. Covariates such as age, sex, race, mechanism of injury, thoracic AIS, SI, GCS on arrival, modified frailty index (mFI-11), pre-injury anticoagulation, and SSRF were included. To assess continuous variables such as ventilator days, hospital LOS, and ICU LOS, linear regression models were used. The same covariates used in logistic regression were used in the linear regression model. Continuous variables were expressed as unstandardized β coefficients (B) with p values, while binary outcomes were reported as adjusted odds ratios (aORs), 95% CIs, and corresponding p values. Adjusted estimates are displayed in a forest plot (Figure S1; Supplemental File 2, Supplemental Digital Content 2, http://links.lww.com/TA/F581). For the SSRF and nonoperative management (NOM) subgroup analysis, the EA versus LA comparison was performed, and the model included age, sex, race, mechanism of injury, thoracic AIS, GCS on arrival, mFI-11, and SI. We also performed a subgroup analysis based on the RA technique used (epidural vs. non-epidural), and the model was adjusted for patient demographics, timing of RA, thoracic AIS, GCS, mFI-11, and mechanism of injury. We calculated the 11-item mFI, which included hypertension, diabetes mellitus, peripheral arterial disease, cerebrovascular accident, myocardial infarction, angina, congestive heart failure, dementia, chronic obstructive pulmonary disorder, functional dependence, and altered sensorium. Each variable was assigned 1 point if present. The total

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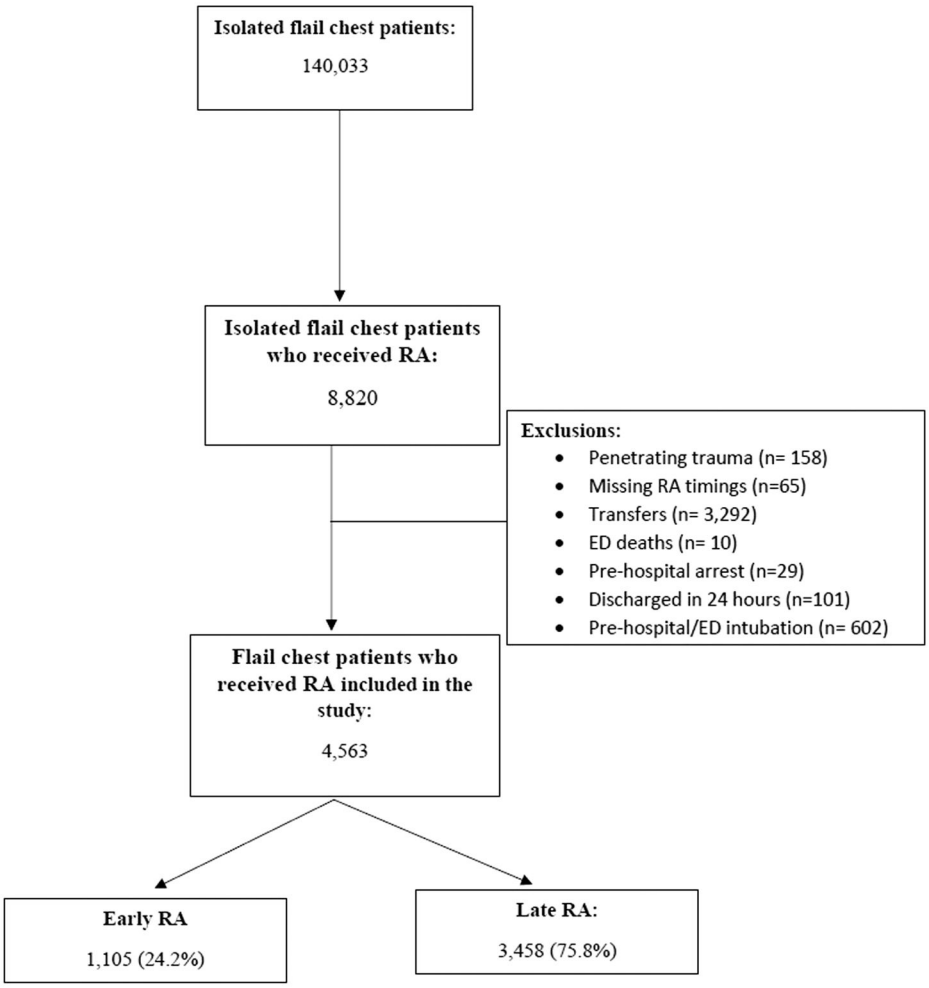


FIGURE 1. Flow chart diagram of the study population.

was divided by 11 to yield a final mFI-11 score ranging from 0.0 to 1.0. Frailty was defined as a modified frailty index ≥ 0.25.[14,15]

The Pearson χ^2 test was used for categorical variables, while nonparametric continuous variables were compared using the Mann-Whitney *U* test. A two-sided *p* value of < 0.05 was considered statistically significant. We performed statistical analysis using the Statistical Package for the Social Sciences (IBM SPSS Statistics for Windows, Version 29.0, IBM Corp).

RESULTS

We identified 4,563 patients (≥ 18 y) with flail chest who received RA. The median age in this cohort was 59 years, and 72.5% were male. The most common mechanism of injury was motor vehicle collision (44.6%) with a median thoracic AIS of 3. Figure 1 illustrates the flowchart diagram used to isolate the study population.

Demographics and Clinical Characteristics

Within the cohort, 24.2% (n = 1,105) received EA, while 75.8% (n = 3,458) underwent LA. The median age and race were similar between the two groups, but the LA group had a higher proportion of females ($p = 0.014$). The LA group had different ED vital signs, including higher HR, lower SBP, and higher SI on arrival ($p < 0.05$). Even though there was statistical significance in HR, SBP, and GCS, the clinical differences were small. Injury Severity Score and mFI-11 were higher in the LA group compared with the EA group ($p < 0.05$). Oxygen saturation at presentation was similar between the groups. Pre-injury anticoagulation was more common in the LA group (8.4% vs. 5.2%, $p < 0.001$). There were also higher rates of SSRF and chest tube placement in the LA cohort ($p < 0.001$) (Table 1).

Outcomes

No difference in mortality was observed between the two groups (aOR, 0.81; 95% CI, 0.49–1.33, $p = 0.397$). In

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TABLE 1. Patient Demographics and Clinical Characteristics

Variables	Early Anesthesia (≤ 24 h) (n = 1,105)	Late Anesthesia (> 24 h) (n = 3,458)	*p*
Age (years)	59 [48–69]	59 [48–69]	0.642
Sex (female), n (%)	272 (24.7)	974 (28.2)	0.059
Race, n (%)			
White	878 (80.2)	2,728 (79.1)	0.717
Black	96 (8.8)	327 (9.5)	
Other	121 (11.1)	393 (11.1)	
Mechanism, n (%)			
MVC	476 (43.1)	1,561 (45.1)	0.079
Falls	367 (33.2)	1,025 (29.6)	
Other	262 (23.7)	872 (25.2)	
Thoracic AIS	3 [3–4]	3 [3–4]	0.352
ISS	17 [14–24]	21 [14–27]	< 0.001
mFI-11	0.09 [0.0–0.09]	0.09 [0.0–0.18]	0.002
ED vital signs			
HR	86 [74–100]	89 [76–103]	< 0.001
SBP	137 ± 27	136 ± 27	0.004
SI	0.62 [0.52–0.75]	0.65 [0.54–0.81]	< 0.001
GCS	15 [15–15]	15 [15–15]	< 0.001
Oxygen saturation	95 [92–98]	95 [92–98]	0.355
Pre-injury anticoagulation, n (%)	58 (5.2)	289 (8.4)	< 0.001
SSRF, n (%)	332 (30.0)	1,525 (44.1)	< 0.001
Chest tube insertion, n (%)	574 (51.9)	2,112 (61.1)	< 0.001

AIS, Abbreviated Injury Scale; ED, emergency department; GCS: Glasgow Coma Scale; HR, heart rate; ISS, Injury Severity Score; mFI-11, modified Frailty Index; MVC, motor vehicle collision; SBP, systolic blood pressure; SI, Shock Index; SSRF, surgical stabilization of rib fractures. Using the χ^2 test and the Mann-Whitney *U* test as appropriate.

TABLE 2. Unadjusted Outcomes Across the Timing of Anesthesia

Variables	Early Anesthesia (≤ 24 h) (n = 1,105)	Late Anesthesia (> 24 h) (n = 3,458)	*p*
Mortality, n (%)	29 (2.6)	88 (2.5)	0.884
In-hospital complications, n (%)			
Unplanned Intubation	57 (5.2)	292 (8.4)	< 0.001
DVT	15 (1.4)	84 (2.4)	0.033
PE	15 (1.4)	54 (1.6)	0.628
ARDS	15 (1.4)	41 (1.2)	0.652
VAP	14 (1.3)	68 (2.0)	0.128
Stroke/CVA	6 (0.5)	13 (0.4)	0.453
Unplanned ICU admission	48 (4.3)	278 (8.0)	< 0.001
Tracheostomy	29 (2.6)	138 (4.0)	0.035
Hospital disposition, n (%)			
Routine discharge	588 (53.2)	1,586 (45.9)	< 0.001
Inpatient rehabilitation	163 (14.8)	664 (19.2)	< 0.001
Skilled nursing facility	110 (10.0)	430 (12.4)	0.026
LTACH	15 (1.4)	69 (2.0)	0.170
Other	189 (17.1)	656 (19.0)	0.164
Hospital LOS	9 [6–13]	11 [8–16]	< 0.001
ICU LOS	4 [3–7]	6 [3–9]	< 0.001
Ventilator days	5 [2–9]	5 [3–11]	0.165

AKI, acute kidney injury; ARDS, acute respiratory distress syndrome; CAUTI: catheter-associated urinary tract infection; CVA, cerebrovascular accident; DVT, deep vein thrombosis; ICU, intensive care unit; LOS, length of stay; LTACH, long-term acute care hospital; MI, myocardial infarction; PE, pulmonary embolism; VAP, ventilator-associated pneumonia. Using the χ^2 test and the Mann-Whitney *U* test as appropriate.

hospital complications such as deep vein thrombosis, pulmonary embolism, acute respiratory distress syndrome, and ventilator-associated pneumonia are comparable between the two groups. In the LA group, there was an increase in unplanned intubation (aOR, 1.52; 95% CI, 1.09–2.12, $p = 0.015$), unplanned ICU admission (aOR, 1.93; 95% CI, 1.34–2.79, $p < 0.001$). There were more unfavorable discharge dispositions in the LA group, with a lower rate of routine discharge (aOR, 0.83; 95% CI, 0.70–0.99, $p = 0.036$) (Figure S1, Supplemental File 2, Supplemental Digital Content 2, http://links.lww.com/TA/F581).

There were also longer hospital LOS ($B = +2.27$ d, $p < 0.001$) and ICU LOS ($B = +1.11$ days, $p < 0.001$) in the LA group, but ventilator days ($B = -0.79$ d, $p = 0.425$) were comparable between the two groups (Tables 2 and 3).

SSRF Subgroup Analysis

A total of 1,857 patients underwent SSRF; 17.9% (n = 332) received EA, and 82.1% (n = 1,525) received LA. Among patients with available procedural timing, SSRF occurred after RA in 46.8% cases, before RA in 8.3%, and concurrently in 44.8%. No surgical site infections were observed in this subgroup. Mortality, unplanned intubation, unplanned ICU admission, and discharge disposition were comparable between the two groups (Figure S1, Supplemental File 2, Supplemental Digital Content 2, http://links.lww.com/TA/F581). LA was associated with longer hospital LOS ($B = +1.69$, $p = 0.032$) and ICU LOS ($B = +1.16$, $p = 0.035$) while ventilator days ($B = +1.89$, $p = 0.200$) were similar between the two groups (Table 4).

NOM Subgroup Analysis

Out of 2,706 patients who were managed nonoperatively, 28.6% (n = 773) received EA and 71.4% (n = 1,933) received LA. Unplanned intubation and mortality were similar in the two groups. LA was associated with higher rates of unplanned ICU admission (aOR, 2.15; 95% CI, 1.33–3.48, $p = 0.002$) and lower odds of routine discharge (aOR, 0.76; 95% CI, 0.62–0.94, $p = 0.009$). There were also increases in hospital LOS ($B = +2.43$, $p < 0.001$) and ICU LOS ($B = +1.03$, $p = 0.002$) in the LA group. Ventilator days were similar in both groups (Table 5).

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TABLE 3. Adjusted Outcomes Across the Timing of Anesthesia

Variables	Adjusted Estimate (aOR/*B*)	95% CI	*p*
Mortality	0.81	0.49–1.33	0.397
In-hospital complications			
Unplanned intubation	1.52	1.09–2.12	0.015
DVT	1.90	0.96–3.77	0.065
PE	0.92	0.46–1.80	0.795
ARDS	0.92	0.46–1.81	0.802
VAP	1.06	0.57–2.01	0.847
Unplanned ICU admission	1.93	1.34–2.79	< 0.001
Tracheostomy	1.21	0.76–1.92	0.430
Hospital disposition			
Routine discharge	0.83	0.70–0.99	0.036
Inpatient rehabilitation	1.19	0.96–1.47	0.108
Skilled nursing facility	1.27	0.97–1.66	0.081
LTACH	1.09	0.58–2.04	0.788
Other	1.12	0.91–1.37	0.288
Hospital LOS	2.27	—	< 0.001
ICU LOS	1.11	—	< 0.001
Ventilator days	0.79	—	0.425

AKI, acute kidney injury; aOR, adjusted odds ratio; ARDS, acute respiratory distress syndrome; *B*, unstandardized beta (in days); CAUTI, catheter-associated urinary tract infection; CVA, cerebrovascular accident; DVT, deep vein thrombosis; ICU, intensive care unit; LOS, length of stay; LTACH: long-term acute care hospital; MI, myocardial infarction; PE, pulmonary embolism; VAP, ventilator-associated pneumonia. Models were adjusted for patient demographics, GCS, SI, SSRF, thoracic AIS, pre-injury anticoagulation, and mechanism of injury.

RA Technique Subgroup Analysis

Out of a total of 4,563 patients who received RA, 47.4% received epidural analgesia, while 52.6% received non-epidural techniques. There were no differences in complications, including unplanned intubation, unplanned ICU admission, unplanned return to the operating room, or mortality, between the two groups. The discharge disposition was also similar between the two groups. Epidural analgesia was not associated with significant differences in hospital LOS, ICU LOS, or ventilator days (Appendix A: Table 2; Supplemental File 2, Supplemental Digital Content 2, http://links.lww.com/TA/F581).

TABLE 4. Adjusted Outcomes in SSRF Subanalysis

Variables	Adjusted Estimate (aOR/*B*)	95% CI	*p*
Mortality	1.74	0.51–5.98	0.377
In-hospital complications			
Unplanned Intubation	1.61	0.93–2.79	0.091
Unplanned return to OR	1.77	0.78–3.91	0.177
Unplanned ICU admission	1.60	0.90–2.84	0.107
Hospital disposition			
Routine discharge	1.03	0.77–1.38	0.848
Inpatient rehabilitation	0.96	0.68–1.37	0.807
Skilled nursing facility	1.02	0.63–1.65	0.936
LTACH	0.90	0.38–2.12	0.814
Other	1.19	0.83–1.71	0.349
Hospital LOS	1.69	—	0.032
ICU LOS	1.16	—	0.035
Ventilator days	1.89	—	0.200

aOR, adjusted odds ratio; *B*, unstandardized beta (in days); ICU, intensive care unit; LOS, length of stay; LTACH, long-term acute care hospital; OR, operating room; SSRF, surgical stabilization of rib fractures. Adjusted *p* values from multivariable logistic and linear regression. Models were adjusted for patient demographics, GCS, SI, mFI-11, thoracic AIS, pre-injury anticoagulation, and mechanism of injury.

TABLE 5. Adjusted Outcomes in Nonoperative Management (NOM) Subanalysis

Variables	Adjusted Estimate (aOR/*B*)	95% CI	*p*
Mortality	0.65	0.37–1.14	0.132
In-hospital complications			
Unplanned Intubation	1.43	0.93–2.18	0.100
Unplanned ICU admission	2.15	1.33–3.48	0.002
Hospital disposition			
Routine discharge	0.76	0.62–0.94	0.009
Inpatient rehabilitation	1.34	1.02–1.74	0.033
Skilled nursing facility	1.36	0.98–1.88	0.064
LTACH	1.31	0.52–3.31	0.565
Other	1.09	0.85–1.39	0.510
Hospital LOS	2.43	—	< 0.001
ICU LOS	1.03	—	0.002
Ventilator days	0.65	—	0.605

aOR, adjusted odds ratio; *B*, unstandardized beta (in days); ICU, intensive care unit; LOS, length of stay; LTACH, long-term acute care hospital; OR, operating room. Adjusted *p* values from multivariable logistic and linear regression. Models were adjusted for patient demographics, GCS, SI, mFI-11, thoracic AIS, pre-injury anticoagulation, and mechanism of injury.

DISCUSSION

In this national cohort of isolated flail chest patients who received RA, EA was associated with reduced rates of in-hospital complications and shorter LOS, and a more favorable discharge disposition. These findings suggest that the timing of RA may highlight an important, potentially modifiable component of the management of flail chest patients. To our knowledge, this is the first national study to evaluate the timing of RA in the isolated flail chest group, addressing a longstanding gap in the literature.

In our study, we observed that EA was associated with decreased unplanned intubation rates and ICU admissions. Pain from injuries related to flail chest causes splinting, resulting in shallow and guarded breathing. This contributes to limited chest expansion, leading to reduced tidal volume, ineffective coughing, retained secretions, and atelectasis.[5] A similar study by Mladenovic et al[11] suggested that blocks placed < 48 hours were associated with decreased respiratory complications and reduced ICU LOS. Our use of ≤ 24 hours as a cutoff for

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EA mirrors the study by Proano-Zamudio et al,[10] who suggested that initiating EA within the first day of hospitalization is associated with decreased unplanned intubation and ICU admissions in older patients with blunt chest wall trauma. Our study applies a similar curve to a high-risk cohort with flail chest and demonstrates that the timing effect persists. Mortality within the cohort of patients with flail chest is associated with factors beyond mere chest wall mechanics, including baseline frailty. This observation explains our finding that the timing of RA does not affect mortality, consistent with other studies.[16,17]

Patients with flail chest are more vulnerable physiologically and could deteriorate early when the pain is not adequately controlled. Poor analgesia could impair the chest wall mechanics, limit secretion clearance, and reduce tidal volume. These factors could increase the risk of infection, respiratory decline, and the need to escalate care. Uhlich and colleagues conducted a study on continuous intercostal nerve blockade for multiple rib fractures and found more hospital-free days with the blockade. Although this study did not focus on block placement timing, it reinforces the effectiveness of RA in reducing LOS.[18] Prior studies on RA techniques suggest that early pain control could reduce complications and LOS.[19–21]. Our study showed that EA was associated with shorter LOS in the main cohort, SSRF, and the NOM subanalysis, suggesting that timing could reduce resource utilization. There was also an improved discharge disposition associated with EA, with a higher proportion of patients being discharged home rather than acute care facilities. This could be due to improved tidal volumes, fewer pulmonary complications, and reduced ICU escalation, allowing patients to regain functional independence early in the hospitalization.[22] There were baseline physiologic differences between the groups with higher HR, lower SBP, and higher SI in the LA group. Although the absolute difference in these variables was modest, the pattern's direction points to greater physiologic stress in the LA cohort. This could be due to the possibility of confounding by indication, where clinicians could have delayed RA administration in those who were less stable or had a high injury burden. Paradoxically, this finding could also be the subgroup of patients who would most likely benefit from timely RA intervention.

In our subgroup analysis of patients who underwent SSRF, EA was associated with lower rates of unplanned intubation and shorter hospital LOS, whereas those managed nonoperatively showed greater benefits with EA. EA was associated with decreased unplanned ICU admissions, increased routine discharges, and decreased hospital and ICU LOS, consistent with previous literature. SSRF improves chest wall mechanics, reduces pain, corrects paradoxical chest wall motion, and leads to better recovery, thereby reducing the relative effect of analgesia timing on outcomes. Patients undergoing SSRF may also receive more protocolized perioperative pain management, which could attenuate the relative effect of RA timing; however, this level of analgesic detail cannot be confirmed in TQIP.[23,24] This pattern reinforces that the impact of EA is most clinically meaningful in patients managed nonoperatively, which is the main cohort for pain control. In our subgroup analysis to compare epidural versus non-epidural RA techniques, there were no differences in in-hospital complications, mortality, discharge disposition, and LOS. Even though thoracic epidural analgesia is historically considered the gold standard for chest wall analgesia, recent studies suggest that alternate regional techniques provide comparable outcomes in patients with rib fractures.[22] Previous studies report similar outcomes between epidural and non-epidural techniques, such as paravertebral, serratus anterior block, and neuraxial analgesia.[25,26] Hence, the type of RA strategy yields comparable outcomes in flail chest; the selection technique should be guided by patient factors and institutional expertise.

This study has several limitations. Although our regression model included multiple covariates, residual confounding may persist due to unmeasured clinical factors. The data in the registry may be subject to selection or misclassification bias. Similar to SSRF, the use of RA in this cohort is subject to treatment-selection bias, since only a small proportion of isolated flail chest patients received RA. Flail chest patients were identified using AIS codes rather than radiologic confirmation, which may not fully capture the severity of displacement of these injuries. TQIP does not include clinical chest examination variables, and we could not distinguish radiographic flail patterns from clinically apparent flail chest. Since the fracture's anatomic location is unavailable in TQIP, we could not determine the fracture site or the number of ribs involved. ACS-TQIP lacks details on the RA received by the patient, whether as a single shot or a catheter-based infusion. Although subcutaneous nerve blocks were excluded to improve the definition of RA, procedural coding in this database does not provide anatomic detail regarding the specific target of the block. Cryoablation could not be reliably identified due to a lack of consistent procedural coding specificity in TQIP and may represent an unmeasured analgesic adjunct, particularly in patients undergoing SSRF. The sequence of clinical events on TQIP could not be ascertained due to the absence of time-stamped data for outcomes such as re-intubation, ICU readmission, initiation or cessation of mechanical ventilation, and transfers from hospital to ICU. Hence, we cannot determine whether these events occurred before or after the RA. We were also unable to report the ventilator-free days, which is an important measure of respiratory recovery. The database also lacks pain-specific variables such as pain scores, RA duration, and multimodal analgesia use. As with other retrospective studies, causality cannot be established. The association between the timing of RA and resource utilization may be influenced by provider preferences and institutional practice variation, which this database cannot capture. These limitations highlight the need for future studies to understand the causal effect of analgesia timing on patients with flail chest.

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CONCLUSIONS

This study demonstrates that early RA in patients with flail chest is associated with lower rates of unplanned intubation and ICU admission, and shorter hospital and ICU lengths of stay. Within the SSRF and NOM subgroup, timely RA is also associated with favorable outcomes. Future prospective studies with more granular clinical data and standardized pain management protocols are necessary to confirm these findings and establish the optimal timing for RA in the treatment of flail chest.

AUTHORSHIP

S.H., J.K., and B.Z. contributed to conception and study design. S.H., K.P., G.F., V.K., M.P., and R.M. were involved in the literature review. S.H., R.M., G.F., J.K., I.S., and M.P. contributed to data acquisition. S.H., B.Z., J.K., I.S., and K.P. contributed to data analysis and interpretation. S.H., V.K., B.Z., J.K., M.P., and R.M. contributed to the drafting of the manuscript. J.K., B.Z., K.P., I.S., and G.F. contributed to critical revision.

DISCLOSURE

Conflicts of Interest: None. All JTACS Disclosure forms have been supplied and are provided as Supplemental Digital Content (http://links.lww.com/TA/F582).

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